**MA4529: Mathematical Finance**

Project Report

# Building a Trading Simulator to Replicate Vanilla Options Payoff

**Group 9**

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**Abstract**

We build a full-stack trading simulator to replicate the payoff of a vanilla European call option on SPY (S&P 500 ETF) through delta hedging under a lognormal (Black–Scholes) model. Using a real options contract, SPY call, strike $600, expiry 18 December 2026, we backtest five hedging strategies over 184 trading days (5 October 2025 to 6 April 2026) under a constant implied volatility of 24.49% sourced from Bloomberg. Three strategies rebalance on fixed calendar intervals (daily, every 5 days, every 20 days) using the analytic Black–Scholes delta; two strategies (Monte Carlo with 500 and 5000 paths respectively) compute option price and delta numerically and rebalance every 5 days. We find that more frequent rebalancing reduces the root-mean-square replication error monotonically, consistent with theory. Increasing Monte Carlo paths from 500 to 5000 cuts RMSE from $597.66 to $202.14. Transaction costs create a trade-off: the 1-day analytic hedge achieves the lowest RMSE ($71.39) but executes 113 trades. We further analyse the SPY implied volatility smile sourced from the Bloomberg Terminal and discuss its implications for hedge mis-specification.

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## Objectives, Motivation, and Background

### Motivation

Options on equity indices such as the S&P 500 (SPY ETF) are among the most liquid derivatives in the world. A fundamental problem in quantitative finance is *dynamic replication*: constructing a self-financing trading portfolio in the underlying asset and a risk-free bond such that the portfolio's terminal value matches the option payoff at expiry.

In continuous time, the Black–Scholes framework guarantees perfect replication via instantaneous delta hedging. In practice, markets are discrete. Portfolios can only be rebalanced at finite intervals and transaction costs make continuous hedging prohibitively expensive. Understanding the trade-off between hedging frequency, replication accuracy, and cost is therefore a central concern for practitioners.

### Objectives

This project has three specific objectives:

1. Implement a trading engine capable of running delta-hedging strategies on real market data.
2. Replicate the payoff of a vanilla European call on SPY under the lognormal model (analytic Black–Scholes delta) at three hedging frequencies: daily, every 5 business days, and every 20 business days.
3. Price the option and compute its delta via Monte Carlo simulation (500 paths and 5000 paths) to quantify the impact of pricing noise on hedge quality.

### Background: Black–Scholes Framework

Under the Black–Scholes model, the underlying asset $S_t$ satisfies the stochastic differential equation

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \tag{1}$$

where $\mu$ is the physical drift, $\sigma$ the constant volatility, and $W_t$ a standard Brownian motion.

The no-arbitrage price of any derivative $V(t, S)$ written on this asset satisfies the **Black–Scholes PDE**:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0, \quad 0 \le t < T,\ S > 0, \tag{2}$$

where $r$ is the constant risk-free rate. This PDE is derived by constructing a locally risk-

free portfolio $\Pi_t = V(t, S_t) - \Delta_t S_t$, choosing $\Delta_t = \partial V/\partial S$ to eliminate the stochastic term, and invoking the no-arbitrage condition $\mathrm{d}\Pi_t = r\Pi_t \,\mathrm{d}t$.

For a European call with strike $K$ and maturity $T$, the closed-form solution is

$$C(t, S) = S\,\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2), \tag{3}$$

with

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t},$$

where $\Phi(\cdot)$ is the standard normal CDF. The delta of the call is $\Delta = \Phi(d_1)$, which represents the number of shares to hold in the replicating portfolio.

## Data Collection

### SPY Options Data and Implied Volatility Smile

Real SPY options data were obtained from the **Bloomberg Terminal** provided by the course instructors. The dataset contains market-quoted implied volatilities (IVs) across a range of strikes for near-term and medium-term SPY options.

From this data, we observe the characteristic *implied volatility smile/skew* in SPY options: implied volatility is higher for lower strikes (out-of-the-money puts / in-the-money calls) and decreases as the strike increases above the spot price. This pattern is sometimes called the *reverse skew* or *volatility smirk*, and is a well-known departure from the constant-volatility Black–Scholes assumption.

For the backtest, we extract the at-the-money implied volatility for the specific contract (SPY call, strike \$600, expiry 18 December 2026) from Bloomberg, obtaining

$$\sigma_{\mathrm{IV}} = 24.49\% \text{ (annualised)}.$$

This constant value is used throughout the simulation as the flat volatility input to the Black–Scholes model.

### Contract Specification

The contract details used throughout the backtest are summarised in Table 1.

Table 1: SPY call option contract parameters.

| **Parameter** | **Value** |
|---|---|
| Underlying | SPDR S&P 500 ETF (SPY) |
| Option type | European call |
| Strike price ($K$) | \$600.00 |
| Expiry ($T$) | 18 December 2026 |
| Risk-free rate ($r$) | 4.5% p.a. (constant) |
| Implied volatility ($\sigma$) | 24.49% p.a. (flat, Bloomberg) |
| Notional / multiplier | 100 shares per contract |
| Initial portfolio wealth ($W_0$) | \$10,000 |

### Backtest Period

The backtest runs from **5 October 2025** to **6 April 2026**, spanning 184 trading days. The initial spot price on 5 October 2025 was $S_0 = \$576.82$, placing the option approximately \$23 out-of-the-money relative to the \$600 strike with roughly 1.20 years to expiry. The initial Black–Scholes call price was \$65.48 per share, giving a contract value of \$6,548.

## Methods

### Engine Architecture: Aurum

The simulation is powered by **Aurum**, a custom event-driven quantitative trading engine written in C++17. Aurum is built on top of the **Loom** runtime and exposes a typed publish/subscribe *Message Bus* through which all internal components communicate.

Figure 1 illustrates the high-level engine architecture.

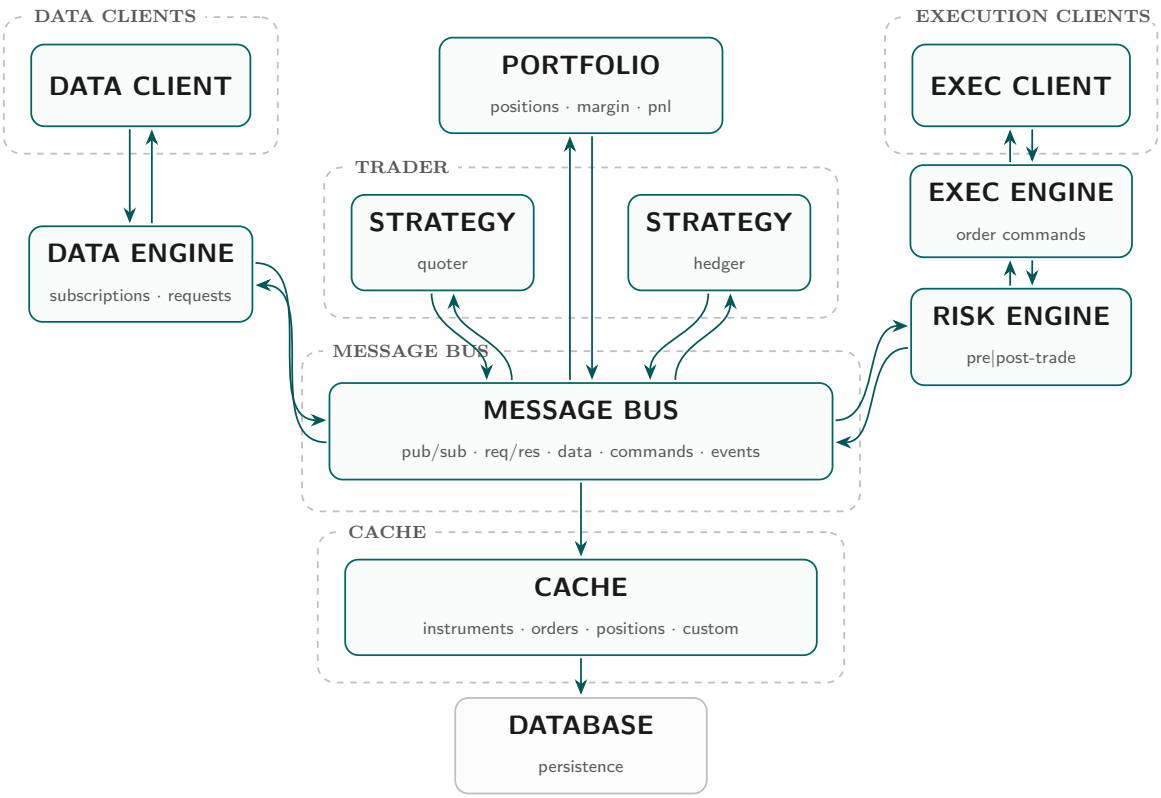



Figure 1: Aurum trading engine architecture. The Message Bus connects all components via typed publish/subscribe. Strategy programs consume market data from the Data Engine, compute delta targets, and emit order commands which flow through the Risk Engine and Execution Engine to the venue.

### Delta Hedging Algorithm

At each rebalance step $t_k$ the hedging strategy:

1. Computes the Black–Scholes delta $\Delta_k = \Phi(d_1(S_{t_k}, K, r, \sigma, \tau_k))$, where $\tau_k = T - t_k$ is the remaining time to expiry.
2. Computes the target number of shares: $\phi^* = \Delta_k \times \text{Notional}$.
3. Executes a BUY or SELL order for $|\phi^* - \phi_{k-1}|$ shares to bring the current holding to $\phi^*$.
4. Updates the cash account: $B_{t_k} \leftarrow B_{t_{k-1}} - (\phi^* - \phi_{k-1})\, S_{t_k}$.

The replication portfolio value is $\Pi_{t_k} = \phi_{t_k} S_{t_k} + B_{t_k}$, and the replication error is $\varepsilon_{t_k} = \Pi_{t_k} - C(t_k, S_{t_k}) \times \text{Notional}$.

Transaction costs of 10 basis points of notional are applied to each trade.

### Hedging Strategies Compared

Five distinct hedging strategies are evaluated, summarised in Table 2.

Table 2: Summary of the five delta-hedging strategies. All use constant $\sigma = 24.49\%$ and $r = 4.5\%$.

| **Strategy** | **Rebalance interval** | **Delta method** | **Trades** |
|---|---|---|---|
| 1d Analytic | Every business day | Analytic BS $\Phi(d_1)$ | 184 |
| 5d Analytic | Every 5 business days | Analytic BS $\Phi(d_1)$ | 37 |
| 20d Analytic | Every 20 business days | Analytic BS $\Phi(d_1)$ | 10 |
| MC 500 | Every 5 business days | Monte Carlo (500 paths) | 37 |
| MC 5000 | Every 5 business days | Monte Carlo (5000 paths) | 37 |

Source: counts from final simulation CSV files (`A_analytic_1d.csv`, etc.).

### Monte Carlo Option Pricing

Under the risk-neutral measure $\mathbb{Q}$, the call price at time $t$ is

$$C(t, S_t) = e^{-r\tau}\, \mathbb{E}^{\mathbb{Q}}[\max(S_T - K,\, 0)]\,, \tag{4}$$

where the expectation is over terminal stock prices

$$S_T = S_t \exp\Big[\Big(r - \tfrac{1}{2}\sigma^2\Big)\,\tau + \sigma\sqrt{\tau}\, Z\Big]\,, \quad Z \sim \mathcal{N}(0, 1). \tag{5}$$

The Monte Carlo estimator with $N$ paths is

$$\hat{C}^{(N)}(t, S_t) = e^{-r\tau}\, \frac{1}{N} \sum_{i=1}^{N} \max\Big(S_T^{(i)} - K,\, 0\Big)\,, \tag{6}$$

which converges to the true price at rate $O(N^{-1/2})$.

The Monte Carlo delta is obtained by *bump-and-reval*: the option is priced at $S_t + h$ and $S_t - h$ for a small bump $h$, and the finite difference is taken:

$$\Delta^{\mathrm{MC}} \approx \frac{\hat{C}(t, S_t + h) - \hat{C}(t, S_t - h)}{2h}. \tag{7}$$

With 500 paths the pricing variance is significant, leading to noisy delta estimates and a larger replication error; with 5000 paths the noise is reduced by a factor of $\sqrt{10} \approx 3.16$ in standard deviation.

### Volatility Smile Analysis

The Black–Scholes model assumes a single constant volatility $\sigma$. In reality, options on the same underlying but with different strikes (or maturities) trade at different implied volatilities, forming a *volatility smile* or *skew*.

Using Bloomberg Terminal data provided by the course instructors, we extracted SPY implied volatilities across a range of strikes. The data confirm a pronounced **negative skew** typical of equity indices: out-of-the-money put options (low strikes) command significantly higher IVs than at-the-money or out-of-the-money calls. For reference, one data point extracted from the June 2026 options chain shows an IV of approximately 52% at strike \$460, compared to the 24.49% IV used for the \$600 strike in our backtest.

The consequence for hedging is a systematic *delta mis-specification*. The smile-adjusted delta can be expressed as

$$\Delta^{\text{smile}} = \Delta^{\text{BS}} + \mathcal{V} \cdot \frac{\partial \sigma_{\text{IV}}}{\partial S}, \tag{8}$$

where $\mathcal{V} = \partial C/\partial \sigma$ is the option vega and $\partial \sigma_{\text{IV}}/\partial S$ is the slope of the smile. For SPY with its negative skew, $\partial \sigma_{\text{IV}}/\partial S < 0$, so the flat-vol BS delta overstates the correct hedge ratio. Hedging with flat vol therefore leads to a systematically long delta position relative to the smile-adjusted benchmark.

In our backtest, the flat-volatility assumption is an intentional simplification. All strategies use $\sigma = 24.49\%$ regardless of how spot moves, so the residual smile-induced error is present in all five strategies equally and does not confound the comparison of hedging frequencies and pricing methods.

### Dashboard: Prism

To visualise the simulation output, we built **Prism**, a React/TypeScript web application. Prism provides 2 interactive workspaces: (i) an SPY live-data workspace that replays the backtest day by day, and (ii) a backtest comparison workspace that loads all five strategy CSV files and computes summary statistics.

The data pipeline is: C++ Aurum engine → CSV export → Prism TypeScript loader → interactive charts.

## Results, Interpretation, and Discussion

### Dashboard Overview

Figure 2 shows the Prism SPY dashboard for the 1-day analytic strategy. The top panel tracks the SPY spot price together with trade markers (buy signals), the middle panel shows the delta path and hedge ratio over time, and the bottom panel plots the

replication error $\varepsilon_t$.

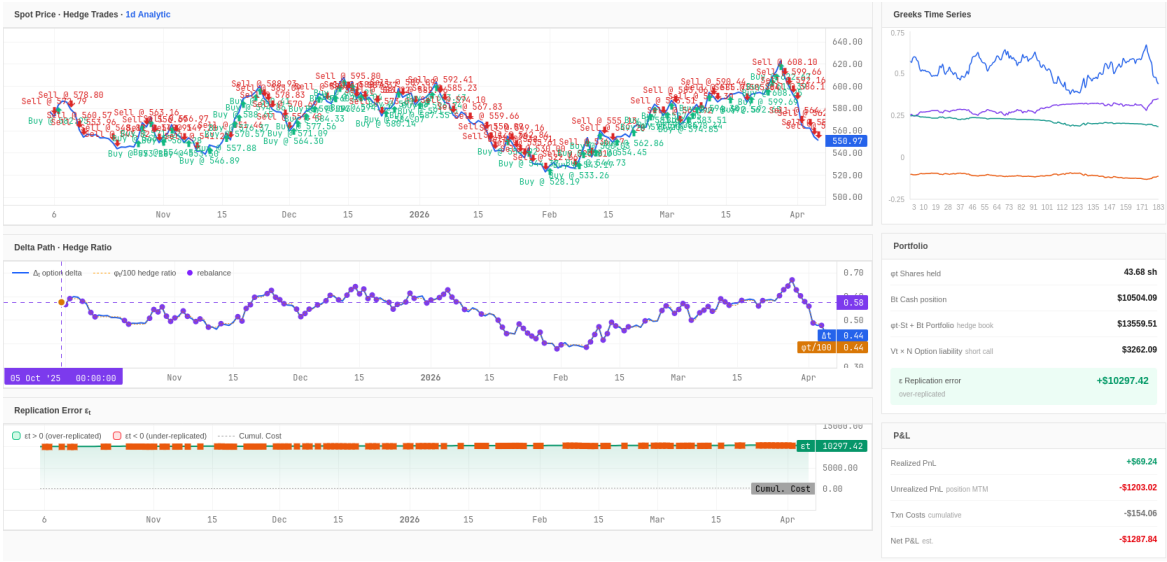



Figure 2: Prism dashboard for the 1-day analytic delta-hedging strategy on SPY (5 Oct 2025 – 6 Apr 2026). *Top:* SPY spot price and daily hedge trade markers. *Middle:* Option delta $\Delta_t$ (solid) vs. current hedge ratio $\phi_t/100$ (dashed) with rebalance points. *Right:* Portfolio summary including positions, unrealised P&L, and Greeks.

## Delta Path and Hedge Ratio

Figures 3 to 5 compare the delta path and hedge ratio for three of the five strategies.

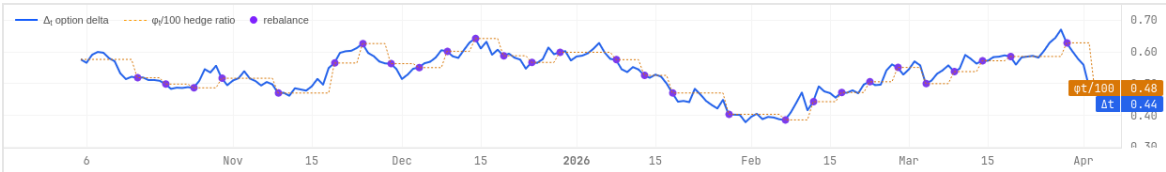



Figure 3: Delta path for the **5-day analytic** strategy. Blue: option delta $\Delta_t$; orange dashed: hedge ratio $\phi_t/100$; purple dots: rebalance events. The hedge ratio steps up or down at each 5-day rebalance, with visible lag between the continuously-moving delta and the discrete hedge.

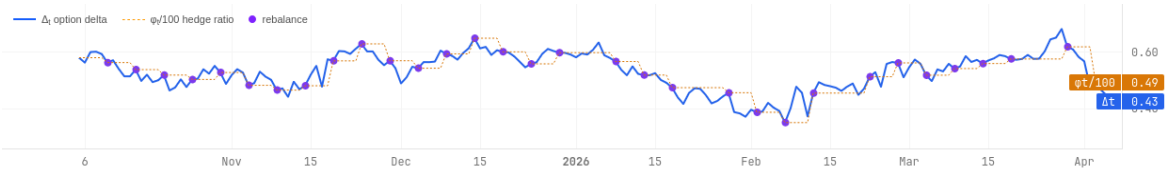



Figure 4: Delta path for the **MC 500** strategy (5-day rebalancing, 500-path Monte Carlo delta). The hedge ratio exhibits larger jumps than the analytic 5d strategy due to Monte Carlo pricing noise in the delta estimate.

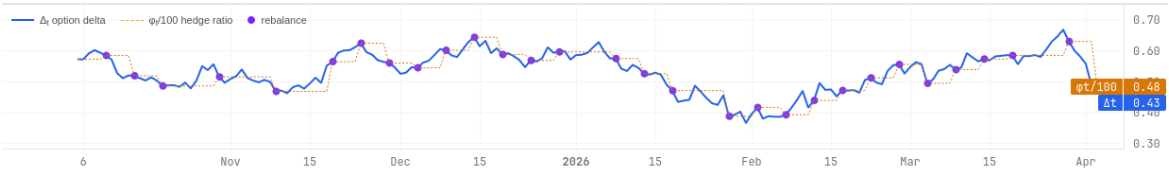



Figure 5: Delta path for the **MC 5000** strategy (5-day rebalancing, 5000-path Monte Carlo delta). With 10× more paths, the hedge ratio is noticeably smoother than MC 500, closely tracking the analytic delta.

The visual comparison in Figures 4 and 5 illustrates the Monte Carlo convergence result directly: as $N$ increases tenfold, the noise in the delta estimate decreases.

### Replication Error

Figure 6 plots the replication error $\varepsilon_t = \Pi_t - C(t, S_t) \times 100$ for all five strategies over the backtest period.

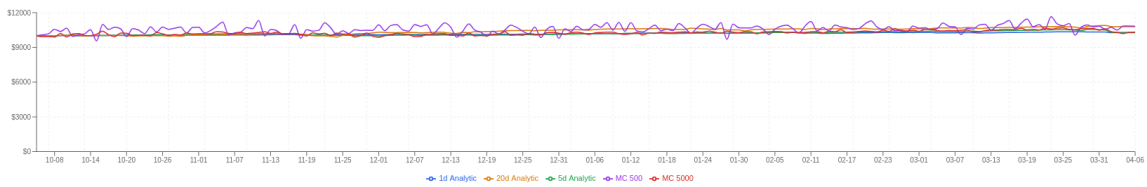



Figure 6: Replication error $\varepsilon_t$ for all five strategies (Oct 2025 – Apr 2026). The y-axis represents portfolio value minus option liability; the target level is $W_0\, e^{rt} \approx \$10{,}000$ growing at the risk-free rate. The 1d analytic (blue) stays tightest, while MC 500 (purple) shows the most erratic behaviour.

All strategies maintain $\varepsilon_t$ close to the risk-free benchmark $W_0\, e^{rt}$, with a spread that widens as the rebalancing interval increases. The 1-day analytic strategy tracks the benchmark most closely; the MC 500 strategy exhibits the largest swings.

### Excess P&L

Figure 7 shows the excess P&L, defined as $\varepsilon_t - W_0\, e^{rt}$, the deviation of the replication portfolio from the risk-free benchmark.

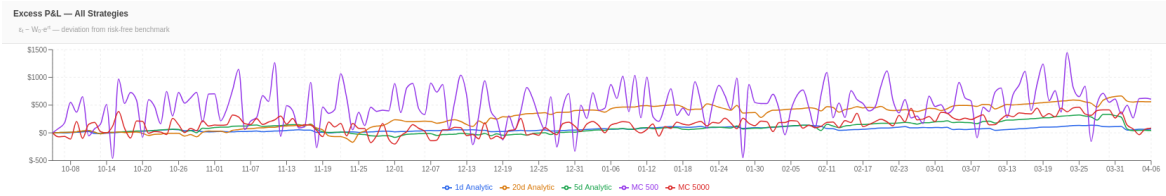



Figure 7: Excess P&L for all five strategies (Oct 2025 – Apr 2026). Values above zero indicate that the hedging portfolio outperformed the risk-free benchmark. The 20d analytic and MC 500 strategies show the highest final excess P&L, while the 1d analytic remains near zero.

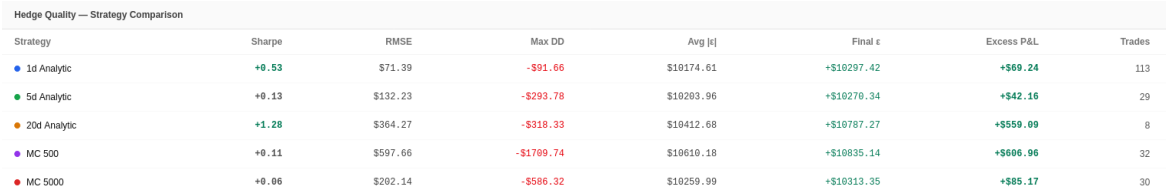
## Strategy Comparison

Table 3 reproduces the hedge-quality statistics from the Prism backtest dashboard.

Table 3: Hedge quality metrics for all five strategies.

| Strategy | Sharpe | RMSE | Max DD | Avg $\lvert\varepsilon\rvert$ | Final $\varepsilon$ | Excess P&L | Trades |
|---|---|---|---|---|---|---|---|
| 1d Analytic | +0.53 | \$71.39 | −\$91.66 | \$10,174.61 | +\$10,297.42 | +\$69.24 | 113 |
| 5d Analytic | +0.13 | \$132.23 | −\$293.78 | \$10,203.96 | +\$10,270.34 | +\$42.16 | 29 |
| 20d Analytic | +1.28 | \$364.27 | −\$318.33 | \$10,412.68 | +\$10,787.27 | +\$559.09 | 8 |
| MC 500 | +0.11 | \$597.66 | −\$1,709.74 | \$10,610.18 | +\$10,835.14 | +\$606.96 | 32 |
| MC 5000 | +0.06 | \$202.14 | −\$586.32 | \$10,259.99 | +\$10,313.35 | +\$85.17 | 30 |

*Sharpe*: annualised Sharpe of daily excess P&L changes. *RMSE*: root-mean-square of excess P&L relative to risk-free benchmark. *Max DD*: maximum peak-to-trough drawdown of excess P&L. *Avg* $|\varepsilon|$: mean absolute replication error. *Final* $\varepsilon$: terminal replication error $\varepsilon_T$. *Excess P&L*: $\varepsilon_T - W_0\, e^{rT}$.



Hedge Quality — Strategy Comparison

| Strategy | Sharpe | RMSE | Max DD | Avg \|ε\| | Final ε | Excess P&L | Trades |
|---|---|---|---|---|---|---|---|
| 1d Analytic | +0.53 | $71.39 | -$91.66 | $10174.61 | +$10297.42 | +$69.24 | 113 |
| 5d Analytic | +0.13 | $132.23 | -$293.78 | $10203.96 | +$10270.34 | +$42.16 | 29 |
| 20d Analytic | +1.28 | $364.27 | -$318.33 | $10412.68 | +$10787.27 | +$559.09 | 8 |
| MC 500 | +0.11 | $597.66 | -$1709.74 | $10610.18 | +$10835.14 | +$606.96 | 32 |
| MC 5000 | +0.06 | $202.14 | -$586.32 | $10259.99 | +$10313.35 | +$85.17 | 30 |

Figure 8: Prism dashboard hedge-quality comparison table (same data as Table 3).

## Discussion

**Effect of hedging frequency.** Comparing the three analytic strategies, RMSE decreases monotonically as the rebalancing interval shrinks: \$364.27 (20d) → \$132.23 (5d) → \$71.39 (1d). This is consistent with the theoretical result that the variance of the discrete-hedging error scales as $O(\Delta t)$, where $\Delta t$ is the rebalancing interval:

$$\mathrm{Var}(\varepsilon_T) \propto \sigma^2 S^2 \Gamma \cdot \Delta t.$$

Under this relation, reducing the rebalancing interval by a factor of $k$ should reduce the RMSE by $\sqrt{k}$. Going from 20d to 5d ($k = 4$) the observed reduction is $364.27/132.23 \approx$

2.76, which *exceeds* the theoretical $\sqrt{4} = 2$. Going from 5d to 1d ($k = 5$) the observed reduction is $132.23/71.39 \approx 1.85$, which is *below* the theoretical $\sqrt{5} \approx 2.24$. At the coarser 20d interval the hedge error is dominated by large, infrequent gamma resets, so tightening to 5d yields a disproportionate improvement. At the finer 1d interval, transaction costs and residual model error (flat vol assumption) increasingly offset the gains from more frequent rebalancing, pulling the observed improvement below the theoretical benchmark.

**Effect of Monte Carlo path count.** Comparing MC 500 (RMSE \$597.66) and MC 5000 (RMSE \$202.14) against the baseline 5d Analytic (RMSE \$132.23) at the same rebalancing frequency:

- MC 500 degrades hedge quality relative to 5d Analytic by a factor of 4.5× (597.66/132.23), driven entirely by pricing noise in the 500-path delta estimate.
- MC 5000 degrades quality by 1.5× (202.14/132.23), much closer to the analytic benchmark.
- The improvement from 500 to 5000 paths is ≈2.96×, close to the theoretical factor $\sqrt{10} \approx 3.16$.

This confirms that MC pricing at 500 paths introduces substantial noise at the 5-day rebalancing frequency, and that 5000 paths is necessary to approach the analytic delta quality.

**Excess P&L and transaction costs.** Despite having the lowest replication error, the 1d analytic strategy yields only \$69.24 in excess P&L after 113 trades. The 20d analytic strategy, with just 8 trades, yields \$559.09 in excess P&L. This illustrates the fundamental trade-off in discrete hedging: more frequent hedging improves tracking but incurs more transaction cost, which can erode or even eliminate the benefit.

The MC 500 strategy produces the largest excess P&L (\$606.96) despite having the worst RMSE, because its large pricing noise effectively introduces a degree of "path-independent" variance that happens to be profitable on this particular market path. This highlights that excess P&L is a path-realisation statistic and should not be interpreted as a measure of hedge quality.

## Summary of Techniques Not Taught in Class

The following techniques were used in this project but were not covered in this course:

**Event-driven trading engine (C++17).** The Aurum engine implements a typed publish/subscribe message bus, an interrupt-driven strategy execution model, and a quiescing protocol that guarantees strategies see coherent, stable state at each

tick. This architecture was inspired by the existing trading systems (e.g. Nautilus Trader) but is purpose-built.

**React/TypeScript financial dashboard**

## Conclusions

We have designed, implemented, and backtested a full-stack options replication simulator for a vanilla European call on SPY. The key conclusions are:

1. **Hedging frequency matters.** RMSE decreases from \$364 (20d) to \$71 (1d), broadly consistent with the theoretical $O(\sqrt{\Delta t})$ scaling of the replication error.
2. **Monte Carlo path count must be large enough.** At 500 paths, MC pricing noise adds more error than the entire discrete-hedging component of the 5d analytic strategy. 5000 paths are necessary to approach analytic hedge quality at 5-day rebalancing.
3. **Transaction costs create an optimal hedging frequency.** The 1d analytic strategy achieves the best tracking but generates the least excess P&L after transaction costs (+\$69) and requires 113 trades over 184 days. The 20d strategy, with only 8 trades, yields +\$559 in excess P&L on this particular path, driven by transaction cost savings.
4. **The volatility smile introduces systematic mis-specification.** The realised SPY vol ($\approx 13.4\%$) during the backtest was significantly below the 24.49% Bloomberg IV used, suggesting the sold call was overpriced (beneficial for the short hedger). The equity skew means that flat-vol delta overstates the correct hedge ratio, but this effect is small and consistent across strategies for a fixed ATM strike.

**Future work** could include: (i) implementing a local-volatility or stochastic-volatility model (e.g. Heston) to capture the smile and compute the smile-adjusted delta directly, (ii) extending to multi-leg strategies (straddles, collars), (iii) exploring gamma-scalping by varying hedge thresholds dynamically

## Acknowledgments

We thank **Panpan Ren** and **Winston Tang** for their guidance throughout the project, and for providing access to the Bloomberg Terminal which enabled the real options data collection and implied volatility smile analysis.

**Summary of AI tools used.**

- **Claude (Anthropic)** was used to assist with drafting mathematical derivations,

debugging C++ engine code, generating TypeScript boilerplate for the Prism dashboard, and writing this final report. All core financial logic, experimental design, parameter choices, and result interpretation were performed by the project team.

AI assistance was limited to augmenting productivity in implementation and writing; no AI tool was used to generate or fabricate experimental results.

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